

9 Coordinate Systems

9.1 Overview

Description	A coordinate system • allows the conversion from WGS 1984 geodetic or cartesian coordinates to, local cartesian, geodetic or grid coordinates and back. • can be attached to jobs. • can be manually defined. • can be computed in the field. • can be directly received from a reference network. Refer to "18.7.1 Configuration of a Rover Real-Time Connection". • can be downloaded to Infinity. • can be uploaded from Infinity.
Using coordinate systems	Coordinate systems are used on the TS and the CS to combine GNSS data with TS data.
	For TS: An attached coordinate system is not used to reduce any measured distance on a TS instrument.
	All GNSS surveyed points are always stored as WGS 1984 geodetic coordinates regardless of the coordinate system being used. Using a different coordinate system converts the coordinates displayed on the panel, but does not convert and restore the coordinate values in the database DBX.
	For TS: Points surveyed with a TS instrument are always stored in local grid coordinates regardless of the coordinate system being used.
	One coordinate system can be attached to a job at one time. This coordinate system remains attached to the job unless it is changed.
Default coordinate systems	The default coordinate system is WGS 1984. It cannot be deleted. It is not possible to create a coordinate system called WGS 1984. Additional default coordinate systems may be available for certain countries.
Active coordinate system	The active coordinate system is the one attached to the job. One coordinate system is always considered as the active coordinate system.
Automatic coordinate system (RTCM transformation parameters)	When Use auto coordinate system is checked in the RTK Rover Wizard, the coordinate system is directly received from the reference network via RTCM correction data. Refer to "18.7.1 Configuration of a Rover Real-Time Connection".

9.2 Accessing Coordinate System Management

Access	1. From the job menu, select View & edit job properties.
--------	---

2. Go to the **Coordinate system**.
3. Open the selectable list for **Coordinate system**.

Coordinate Systems

Listed are all coordinate systems stored in the database DBX. Any unavailable information is shown as -----.

Coordinate Systems		
Hz 0.0003 g V 0.0001 g @ 13:33		
<None>		
Type -----	Ht mode -----	Design pts -----
CH1903		
Type Classic 3D	Ht mode Orthometric	Design pts User entered
crd_sys1		
Type Classic 3D	Ht mode Ellipsoidal	Design pts User entered

Fn	OK	New	Edit	Delete	More	Fn
----	----	-----	------	--------	------	----

Key	Description
OK	To select the highlighted coordinate system and to return to the previous panel. The selected coordinate system is attached to the job.
New	To create a coordinate system manually. Refer to "9.3 Coordinate Systems - Creating and Editing".
Edit	To edit the highlighted coordinate system. Refer to "9.3 Coordinate Systems - Creating and Editing".
Delete	To delete the highlighted coordinate system. Deletion is not possible if the highlighted coordinate system is active and its source is RTCM.
More	To display information about the type of transformation used, the type of heights computed, the number of control points used for the determination and the date of when the coordinate system was created.
Fn Set deflt	To turn the highlighted coordinate system into a user-defined default coordinate system stored in the instrument.
Fn Default	To recall the deleted default coordinate systems.

9.3

Coordinate Systems - Creating and Editing



Coordinate systems can be defined by manual creation or determined by calculation.

In this chapter, the manual creation of coordinate systems is explained. Refer to "39 Determine Coordinate System" for information on the determination by calculation.



Coordinate systems with a Classic 3D transformation can be defined by manual creation.



The type of transformation of the selected coordinate system determines which elements of a coordinate system can be edited. The name of the

coordinate system, the method of residual distribution and the geoid model in use are always editable.



For coordinate systems with source RTCM, only the geoid model in use can be changed.

However, if no projection is received with the automatic coordinate system, then the projection can also be defined.

Access

In **Coordinate Systems**, highlight a coordinate system. A copy of this coordinate system is taken for further configurations. Press **New** or **Edit**.

New Coordinate System or Edit Coordinate System

New Coordinate System

Hz 0.0000 g
V 0.0002 g

Name: -----

Transformation: <None>

Ellipsoid: WGS 1984

Projection: <None>

Geoid model: <None>

CSCS model: <None>

A coordinate system allows GS & TS measured points to be used together in the same job

Store

Key	Description
Store	To store the coordinate system.

Description of fields

Field	Option	Description
Name	Editable field	A unique name for the new coordinate system. The name can be up to 16 characters long and include spaces.
Residuals		Available for transformations with control points. Manually entered transformations do not have control points. The method by which residuals are distributed throughout the transformation area. The transformation results become more realistic and any strain is dispersed in the transformation.
	1/distance, 1/distance ² , 1/distance ^{3/2}	Distribute the residuals of the control points according to the distance between each control point and the newly transformed point.
	Multiquadratic	Distributes the residuals using a multi-quadratic interpolation approach.
Transformation	Selectable list	The type of transformation.